Live on Cloud Run · Google Cloud AI Builder Cup 2026

Naqsha نقشہ

Walk through your home before the first brick.

From one message about a plot to an LDA-checked floor plan and a 3D home the family can walk through, share, comment on and change.

Gemini 4 Argon designs the house in code and reviews its own drawings. Google ADK agents take the brief and make the revisions. All on Google Cloud.

naqsha-242926648187.us-central1.run.app

github.com/abdm693/hack2skill-hackathon

[Team name] · Lahore first · LDA rules of 9 Feb 2026



In Lahore, a house starts as a sketch and a guess

Johar Town · 10 marla (35′ × 65′, about 2,275 sq ft)

A family of six wants four bedrooms on one budget. They start with a paper sketch and a contractor's guess, then weeks of back-and-forth with an architect.

Nobody walks through it until it's built.

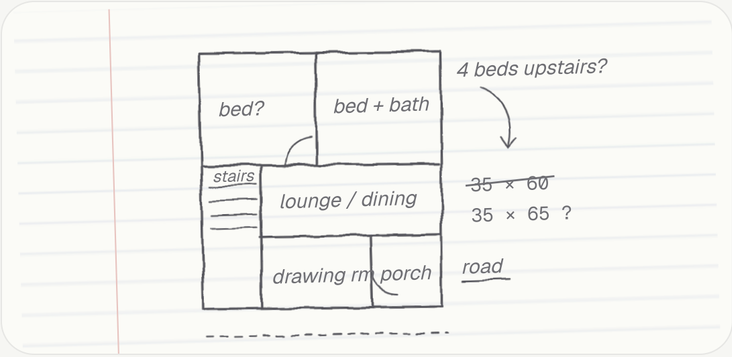
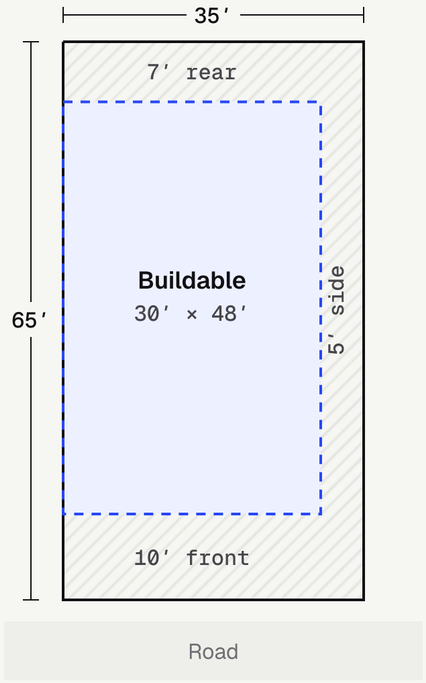


Illustration: how many plans begin, on paper.



Same plot, by the rules.

What that plot must respect under the Lahore Development Authority's (LDA) rules of 9 Feb 2026

10′ · 7′ · 5′	open at the front, the rear and one side
70%	ground coverage, at most
45′	to the top of the parapet
3 storeys	down from 4 in the 2020 text
PCATP stamp	an architect of the Pakistan Council of Architects and Town Planners signs it
Rs 15 / sq ft	scrutiny fee, paid again on every revised plan

What a mistake costs on this plot · worked out from LDA's fee schedule

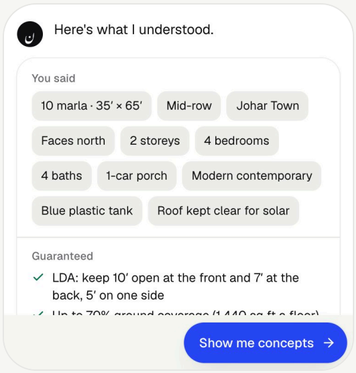
≈ Rs 41,000	for each resubmission of the 2,762 sq ft house Naqsha designed for this plot
Rs 14.5–29k	penalty for its 14′6″ porch pushed 4 ft into the front open space: 58 sq ft at Rs 250 or Rs 500

The rules are written down. They just aren't in the room when the first sketch is drawn.

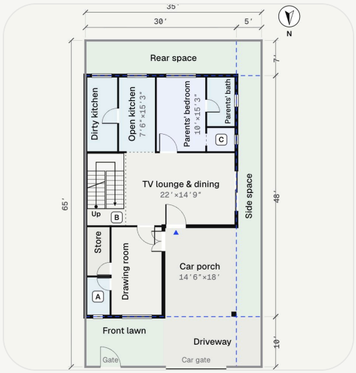
Source: LDA Building & Zoning Regulations 2019, consolidated text of 9 Feb 2026, Appendix C

One sitting, from an idea to a house you can walk through

Brief, concepts, plan, house, walk and revise.
Shown on the live 10 marla sample.











Car porch, ground floor

“Can we extend the car porch 4 ft forward into the front lawn?”

● Not possible

“LDA regulations require a mandatory 10 ft front open setback and do not allow the car porch slab to project into the front lawn…”

~5-6 min: plan, checks, review, 3D

v2, v3 …

Brief

`1 message`

Describe the plot in one message. Naqsha says what it understood and asks only what's missing.

`gemini-3.8-flash on ADK + LDA envelope`

Concepts

`45–94 s`

Four distinct exteriors from the same camera and light, by day and by night.

`gemini-3.1-flash-image × 4, with fallbacks`

Plan

Checked against LDA and livable-home rules. Argon reviews a drawing of it and fixes what fails.

`gemini-4-argon + LDA and livable rules`

House

Argon writes the house as code: structure, facade, interiors, furniture, landscape and lights.

`gemini-4-argon → House SDK code`

Walk

`live`

Mouse look, floor moves and a room navigator. Pin a note on any surface.

`three.js + three-mesh-bvh`

Revise

`~70 s`

A new version with what changed, or the rule that says no and an alternative.

`Google ADK agent, 9 tools`

Times measured on the live Cloud Run service, revision naqsha-00009, 17 Oct 2026 · sample homes open instantly

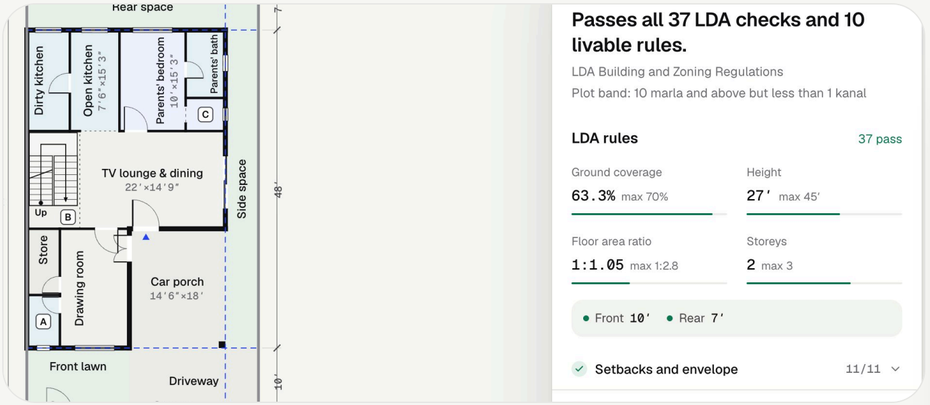
It's live on Cloud Run today

Open a sample home and walk it in a minute.

naqsha-242926648187
.us-central1.run.app



House. The 1 kanal villa: tiled roofs, rooftop solar and a two-car porch.



Plan. The 10 marla sample: 37/37 LDA and 10/10 livable, no warnings.

Enlarged the master bedroom front glazed sliding door/window from 7 x 7.5 ft to 9 x 8.5 ft across nearly the full wall width.

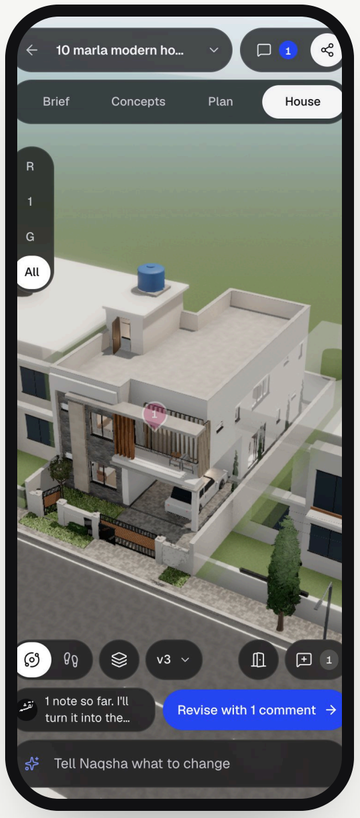
Door 7' → **9' wide** · master bedroom to master balcony

From 1 comment

Widened the master bedroom curtains and opened up the balcony louver spacing to let in more morning sunlight.

The furniture in the master bedroom, the upper screen on the facade redrawn in 3D

Versions. Version 2, from two comments: what changed, in feet.



Phone. The same house.

Argon designs in code, not pixels

So the house is exact, checkable, editable one section at a time, and walkable.

Brief
+ the chosen concept

1 Plan — gemini-4-argon

A schema-bound JSON plan in feet: rooms, doors, windows, stairs, roof and tank. Then Argon reviews a drawing of its own plan.

check

fix: 3 rounds, up to 5

Rules engine — deterministic

18 LDA check types per room and stair, plus 10 livable-home rules

2 Build — design.js · House SDK

Inputs: plan JSON · concept image · SDK types

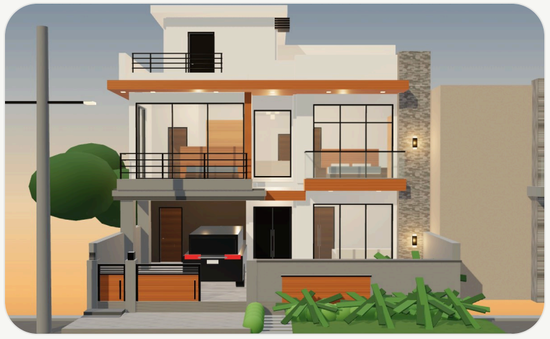
Argon writes 23–25 named sections against our SDK: 44 PBR presets, 57 furniture types, real wall openings, and raw three.js where a detail is bespoke.

structure · facade_stone_pier · facade_left_bay · interior_ground · furnish_g_lounge · landscape · lighting · … each tagged {layer, level, room}

3 Stream

renders after every section





Before: one shot, raw three.js. 1,780 lines in one block.



After: House SDK, in sections. The outlined pier is one section, `facade_stone_pier`.

Same model, two ways of working

481 s — for the one shot: raw three.js, nothing to see until the end, and a blocky house

~5–6 min — today for a 10 marla home: the plan, the checks, Argon's self-review and a high-fidelity house

The whole pipeline now finishes sooner than the one shot took to write blocky code alone.

One-shot spike: 16 Oct 2026, re-shot from a matching front camera · both are 10-marla Lahore houses by Argon

Argon is creative. The rules are code.

Deterministic engines check every plan for LDA and livable-home rules, and Argon reviews its own drawings like an architect.

Rules engine · Live

37/37 LDA · 10/10 livable

on the 10 marla sample, with no warnings. The 1 kanal villa passes 43/43 and 10/10.

18 LDA check types, applied per room and per stair: setbacks, coverage, FAR, height, storeys, 8% daylight and stairs. **10 livable-home rules** on top.

Argon's fix loop on a big 1 kanal brief, measured live

15 → 5 → 0

failures · attempt 1: 15 · attempt 2: 5 · attempt 3: 0

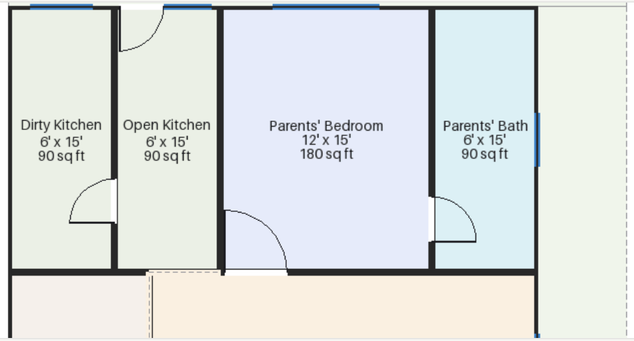
Adaptive: a big brief gets up to 5 rounds while each one still improves the plan.

Livable-home rules, in the live panel · Live

- **Every room has a proper way in**
  Every room is reached through circulation, never through a bedroom or bath, and the family reaches the lounge without crossing the drawing room.
- **Stairs reach every floor and the roof**
  One stacked stair climbs to every floor and the roof, with a 3 ft landing at the foot and the top.
- **The roof door is at the top of the stair**
  The stair tower covers the stair and its top landing, and its door opens beside that landing.
- **Doors and windows fit their walls**
  Every door and window sits on a wall the two rooms share, on the same floor, and fits inside it.

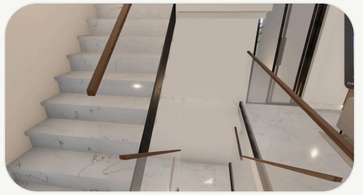
Why deterministic: we audited three houses (two by Argon, one hand-made) and all three had stair-safety defects, one a 22 ft drop with no railing. LDA doesn't define guards, so we apply IBC 2021: 42″ wherever the drop is over 30″.

Argon's self-review · Live



Argon looks at its own floors, drawn like this, and critiques circulation, privacy, daylight and the brief. Then it revises, before the family sees the plan.

Built in 3D, whatever Argon writes



railings on every edge

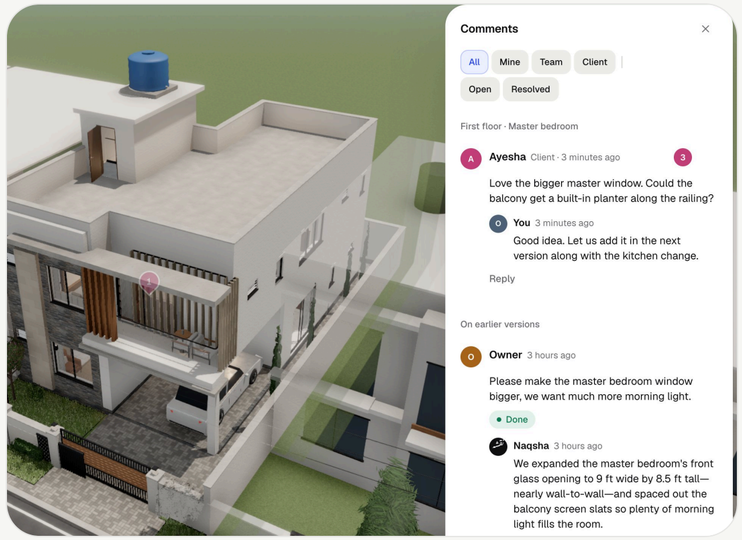


roof door at the top landing

Naqsha is a pre-check, not an approval. A PCATP-registered architect still stamps the plan.

Comment in 3D. The agent revises, or tells you which rule says no.

An agent, not one prompt: it reads only what it needs, dry-runs every layout change against the rules, and saves nothing until it finishes.



What the agent changed · 10 marla, v1 → v2

plan	f_master_slide · 7×7.5 → 9×8.5 ft
code	furnish_f_master · wider curtains
code	facade_upper_screen · louvers opened up
check	36/36 LDA · 12/12 livable · still pass

Google ADK 2.12 · gemini-4-argon

Look
get_rooms · read_sections
list_sections · sdk_reference

↓

Test
check_plan_ops
a dry run on LDA and livable rules

↓

Change only what's needed
apply_plan_ops
write_section · remove_section

↓

Explain
finish
one answer per comment

↑ back to Look for the next comment

The same agent takes requests typed in the Ask bar. If a run fails, a single-shot reviser takes over.

10 marla sample · v1 → v2 · 2 comments — 70 s

First floor · master bedroom window — Done

“Please make the master bedroom window bigger, we want much more morning light.”

“We expanded the master bedroom's front glass opening to 9 ft wide by 8.5 ft tall—nearly wall-to-wall—and spaced out the balcony screen slats so plenty of morning light fills the room.”

Ground floor · car porch — Not possible

“Can we extend the car porch 4 ft forward into the front lawn?”

“LDA regulations require a mandatory 10 ft front open setback and do not allow the car porch slab to project into the front lawn or driveway. Instead, your porch already includes the maximum permitted 2.6 ft cantilevered canopy overhang at the front, and if you need deeper covered parking we could push the lounge wall back slightly inside the building line.”

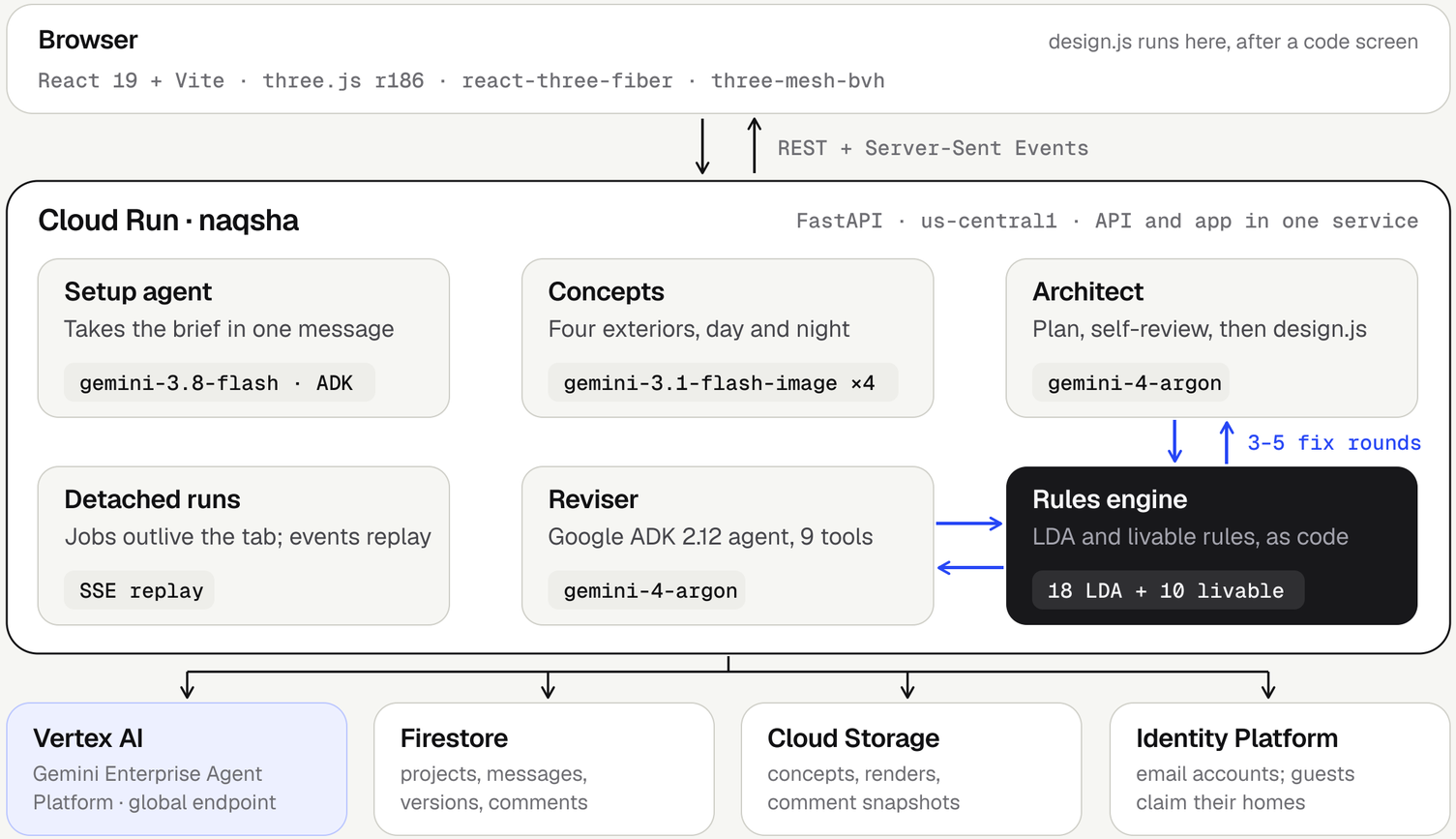
The same porch request on the 1 kanal villa

Ground floor · car porch — Not possible

“The car porch and entrance veranda already sit right on the required 10 ft LDA front setback line… we could instead add an overhead front canopy projecting up to 3 ft past the setback line.”

Answers verbatim from the live samples' ADK runs · left: the live comments sheet (the client note was added for the capture)

One Cloud Run service, Gemini on Vertex AI, state in Firestore



Gemini, matched to each job

Setup `gemini-3.8-flash · ADK`

A conversational agent with 5 tools. Fast and cheap.

Concepts `gemini-3.1-flash-image`

Four at once, by day and night. Falls back to Nano Banana 2.1, then 3 Pro Image.

Plan `gemini-4-argon`

JSON in feet against hard limits, then a review of its own drawings.

Build `gemini-4-argon`

Long, correct code against our typed SDK, from the concept.

Revise `gemini-4-argon + ADK`

Nine tools, a dry run before every write, one save at the end.

One service, one URL. No CORS, and a deploy is one `gcloud run deploy --source .`

google-genai 2.31 · google-adk 2.12 · Cloud Run revision naqsha-00009, 3,600 s request timeout

Built for real quotas, tested by agents

Every step of a run has a guardrail, so a long agent job survives quota errors, bad output and closed tabs. Then agents tested the live app until it held.

1 **Gemini call**
Vertex AI · global

2 **Stream**
SSE to the browser

3 **Code screen**
before a section runs

4 **Browser run**
three.js r186

5 **As-built check**
3D model vs. LDA

6 **Revise**
Google ADK 2.12

Backoff and fallbacks

429s and 5xx errors retry with backoff and jitter, 4 s to 60 s. Concept images fall back to Nano Banana 2.1, then 3 Pro Image.

Degenerate-decoding watch

A runaway tail such as `35.0000000…` aborts and retries the attempt. Prompts cap numbers at 2 decimals.

21 banned patterns

Network, eval, storage, timers and globals are screened out of generated code before it runs.

Self-healing sections

A section that throws is rewritten on its own by Argon. If an ADK run fails, a single-shot reviser takes over.

Engine invariants

Railings at every drop, door keep-outs, the roof door at the landing, and nothing in a setback beyond the chajja allowance.

Detached runs

A design keeps going when the tab closes. When it comes back, every event replays, then the run goes live.

Quality loop Tester agents used the deployed app the way families, architects and builders would. Every finding was triaged against the code, fixed and redeployed.

Round 1 8 testers → **Round 2** 8 testers → **Round 3** 8 testers → **Final check** the must-fix list

each round: test the live app → triage against the code → fix → deploy

~200 fixes merged

123 API tests

All must-fix items confirmed on deployed code

Guardrails: api/app/gemini.py, llm.py, runs.py, web/src/engine · Testing: docs/review/*-triage.md, api/tests (pytest)

Two briefs, two houses, one pipeline, live today

Both samples pass every LDA and livable rule with no warnings. And a short, concrete path from one city to many.



10 marla modern, Johar Town

35′ × 65′ · faces north · 2 storeys · 4 bedrooms

Checks	37/37 LDA · 10/10 livable
Covered area	2,762 sq ft on 3 levels
Revisions	v2 in 70 s · v3 from the Ask bar
Design time	~5-6 min for a 10 marla home



1 kanal Spanish villa

50′ × 90′ · faces south · 2 storeys · 5 bedrooms

Checks	43/43 LDA · 10/10 livable
Roof	solar panels, stair tower
Revisions	porch refused, canopy offered
Big briefs	~6.5 min · fails 15 → 5 → 0

How it scales

Cloud Run, as deployed

Autoscales to 5 instances today, each 2 vCPU and 2 GiB with 40 concurrent requests, CPU always on. The cap is one flag.

A shared Gemini quota

Calls back off on 429s, so a burst queues instead of failing. Four concepts take 45–94 s, with two fallback image models.

Long jobs

Today: an in-memory event log per instance, held by session affinity. Next: the log in Firestore with Cloud Tasks, so any instance can resume.

Cost follows intent

Samples open with no model calls. Briefs and concepts use Flash models; Argon runs only on a full design.

New cities are data

Rules live in one JSON file per authority, and DHA's height and storey caps already switch on.

Evidence: the live samples (revision naqsha-00009) and Cloud Run settings, read 17 Oct 2026 · times measured live

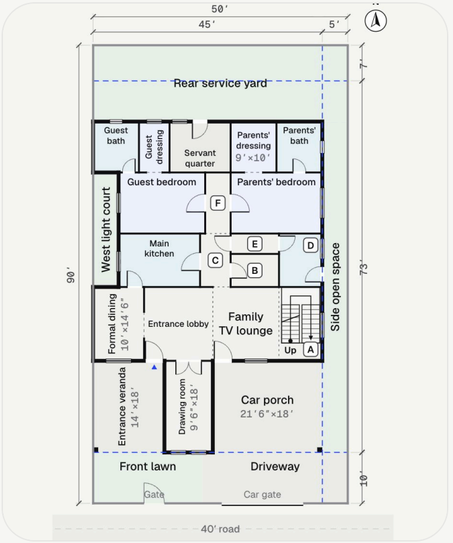
Built for this plot, in this city

The same 1-kanal brief, 50 ft wide. Maket's own FAQ says its designs aren't ready to build; Naqsha checks every plan against LDA rules first.

	Status quo	maket.ai, 16 Oct 2026	Naqsha
Knows the plot	From memory	No: it drew a house wider than the plot	Envelope first; LDA and livable rules on every plan
Local rooms	Yes	US-centric: no drawing room or servant quarter	Drawing room, dirty kitchen, servant quarter
Floors, stairs	On paper	One floor when free; no stairs anywhere	Up to 3 storeys, checked stairs, a mumty
3D	None until built	White-box massing; a walk through walls	Facade, interiors, furniture as code; collisions
Changes	Redraw by hand	Chat edits at 5 credits each, paid plans	Pin a comment; the agent revises or cites the rule
Sharing	In person	A view-only link	Team and client roles; a client link; threads
Speed	Weeks to a drawing	About 6 min to a 2D plan	About 5–6 min to a walkable, checked 3D house



maket.ai: a 59′6″ house for a 50 ft plot, with no warning.

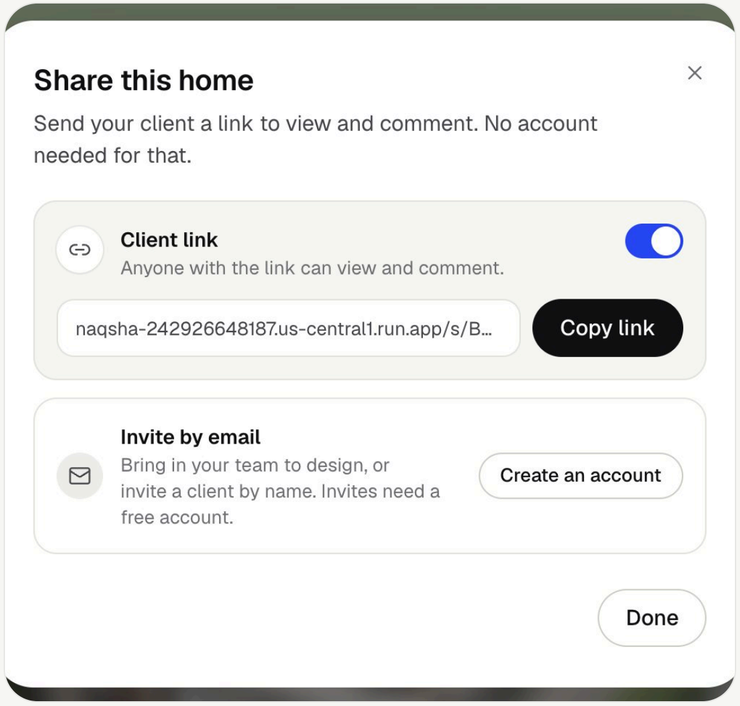


Naqsha: inside the 10′ front, 7′ rear and 5′ side setbacks.

Source: our maket.ai teardown on its free tier, 16 Oct 2026

The family, the architect and the builder, in one home

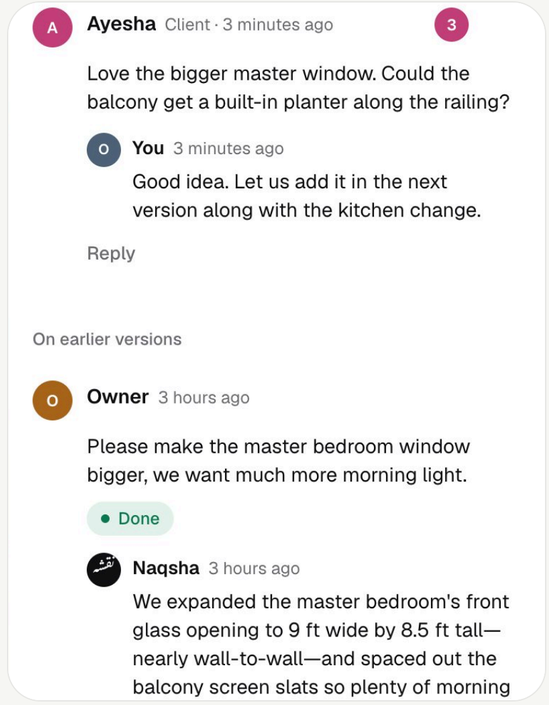
Owner, Team and Client roles. Clients need no account, and the owner picks which comments go into the next version.



Share. A client link that needs no account, or an email invite to a free account.



The client's view at /s/<token>: “Shared with you · You can comment”. They walk it, leave notes and approve v3.



Threads. Every note has a name and a role. Naqsha answers what it changed.

Roles: Owner and Team edit and revise · Client views, comments and approves · maket.ai offers a view-only link (tested 16 Oct 2026)

Free to imagine. Paid to build. Lahore first, then Punjab.

One engine for four audiences. A new authority is a rule file, not a rebuild.

Families	See, walk and agree on the house before paying for drawings, and catch rule breaks before a resubmission that repeats a ≈ Rs 41,000 scrutiny fee.
Architects and builders	Win the client in the first meeting with a walkable concept and an LDA pre-check, then share a link the family can comment on.
Developers and schemes	A sample home for every plot size in a scheme, to sell plots with.
PCATP architects, as stamping partners	They review, stamp and submit a pre-checked plan, printed at 1:100, the scale LDA asks for, with a cover and a compliance report.

How it grows across Punjab · rules are data



An authority switch is built in: DHA height and storey caps today.

Pricing idea, to test in pilots · no prices set yet

Free

The brief, the LDA envelope and four concepts.

free

Per home

Plan, 3D, walkthrough, revisions and a print set.

paid by the family

Pro seats

Client links, branded title blocks, many homes, and a referral to a stamping architect.

per seat, by the firm

For reference, maket.ai charges $20 a month (Homeowner) and $100 a month (Pro).

Pilot metrics to track: first-submission acceptance, days to a first drawing, revision rounds

Every family in Punjab walks through a legal home before paying for drawings

Now

live on revision naqsha-00009

Livable-home rules and Argon's self-review	● Live
Roles, client links and threaded comments	● Live
Accounts; guests claim their homes	● Live
One-message setup agent on ADK	● Live
Day and night concepts	● Live
Print or save as PDF at 1:100	● Live

Milestone

Every must-fix item confirmed on the deployed code.

Next

depth and reach

- Drag-to-swap rooms with an instant LDA re-check
- Google sign-in
- A sandboxed iframe for generated code
- Run events in Firestore, resumable anywhere
- Rule packs for RDA, FDA and GDA, then DHA and Bahria Town

Milestone

Rule packs live for three more authorities.

Later

ideas, not yet specified

- Pilots with LDA-enlisted PCATP architects
- Submission-ready sets for LDA's online system
- Structural and services drafts
- An Urdu interface
- Cost estimates

Milestone

A first pilot with practising architects.

Status: ● Live on the service today, verified 17 Oct 2026 · Next and Later are plans, not commitments

Naqsha نقشہ

Walk through your home before the first brick.

Naqsha is live. Open a sample home in a minute, or design your own in about six.

It said no to the porch, and told us which rule.

Live	naqsha-242926648187.us-central1.run.app
Code	github.com/abdm693/hack2skill-hackathon
Demo	[video URL, under 3 min]

What we'd build next on Google Cloud

- Run events in Firestore with Cloud Tasks, so any instance can resume a design
- Rule packs for RDA, FDA and GDA, then DHA and Bahria Town
- A pilot with LDA-enlisted PCATP architects and Lahore builders

photo photo photo

[Team name]

[Name, role] · [Name, role] · [Name, role]

